

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0011

Gardner 230 kV
May 19, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0011. The Interconnection Customer proposes a 45.3 MW generating facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on February 18, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 8 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$22,178,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: 43Rd St Terminal to Grand Ridge Battery Projects 230 kV
Contingency: 43Rd St Terminal to Grand Ridge Battery Projects 230 kV Circuit 1
Loading: 100.2% (Rating: 843 MVA)
DFAX: 14.7%
Pre-project loading: 99.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
43Rd St Terminal to Grand Ridge Battery Projects 230 kV	Reconductor line	\$39,374,000	56% (\$22,178,000)
	Total allocated to project		\$22,178,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.